VEKTOR RESEARCH SERIES · 2026

Quantitative Portfolio Construction

for Single-Country Equity Markets

A replicable five-stage framework — universe screening, 10,000 portfolio configurations, and benchmark validation.

For professional and institutional investors only

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ABSTRACT

This paper describes the methodology the Vektor platform uses — not the approach we are building toward, but the framework running today. It presents a replicable, quantitative framework for constructing equity portfolios in domestic markets characterised by a bounded investable universe, concentrated sector exposure, and structural liquidity constraints. Drawing on a worked implementation across SGX-listed equities, we demonstrate how disciplined universe screening, 10,000 portfolio configurations mapping the efficient frontier, and index-relative tracking error minimisation can be combined into a coherent, scalable investment process. The principles generalise across any single-country equity market where a national benchmark index exists.

KEY TAKEAWAYS

— A five-stage, benchmark-aware workflow can produce diversified domestic equity portfolios even in small, concentrated markets.

— Liquidity and dividend-continuity screens reduce implementation risk without turning the process into a pure yield strategy.

— 10,000 portfolio configurations generate the empirical efficient frontier; Modern Portfolio Theory identifies the maximum Sharpe portfolio from within it — providing a transparent, reproducible audit trail from universe to final holdings.

1. INTRODUCTION

Why domestic quant mandates are harder than they look: a small universe, sector concentration, and liquidity constraints force discipline in screening and portfolio construction.

This paper describes a five-stage framework: universe definition, liquidity and quality filtering, disciplined universe screening, 10,000 portfolio configurations mapping the efficient frontier, and index-relative tracking error minimisation against the domestic benchmark. We use SGX-listed equities as the primary illustrative dataset, with the Straits Times Index (STI) as the reference benchmark.

> From an implementation perspective, the framework aligns with a standard client-portfolio workflow: a research function defines the investable universe; a portfolio management function approves the strategy and generates target allocations; and an operations function ensures funding is available. This structure supports an auditable mapping from analytical inputs to investable outputs — the same structure embodied in the Vektor platform.

2. UNIVERSE DEFINITION AND SECTOR MAPPING

Define the investable set first: exclude non-ordinary listings, map sectors consistently, and make explicit which parts of the market you are structurally choosing to own — or not own.

Universe definition is not merely administrative. The choice of sectors establishes the strategy's factor exposures from the outset. A strategy that excludes REITs, for example, will exhibit materially different yield and duration characteristics than one that includes them.

> SGX implementation: the starting point is the set of listed ordinary equities — several hundred names. Prior to applying investability constraints, constituents are mapped into broad sector groupings (financials, real estate, industrials) so that sector exposures can be monitored explicitly throughout the optimisation process.

3. LIQUIDITY AND QUALITY FILTERING

Screens should reduce implementation risk: calibrate liquidity to strategy size and use dividend continuity as a practical proxy for business quality.

The quality filter applies dividend continuity as a proxy for business stability, retaining only companies that have paid dividends in at least two of the preceding three fiscal years. This is not a yield-maximisation screen

— it is a quality screen. Companies with consistent dividend histories tend to exhibit lower earnings volatility and more disciplined capital allocation. Liquidity thresholds should be stated explicitly and calibrated to mandate size.

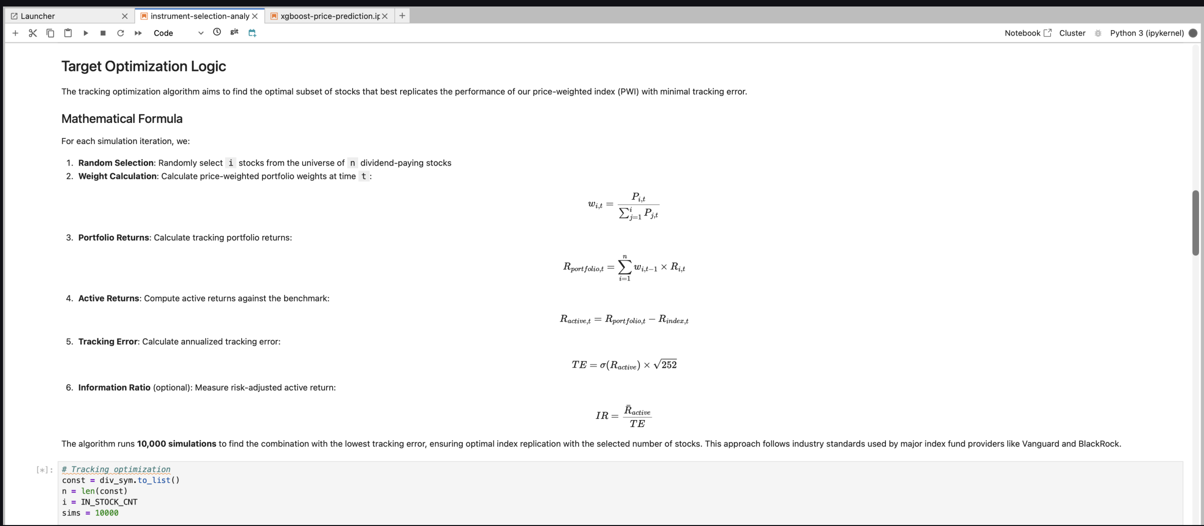
A minimum average daily volume screen (e.g., >100,000 shares/day) reduces implementation shortfall and mitigates the risk of selecting statistically attractive but illiquid securities.

4. PORTFOLIO CONFIGURATION AND EFFICIENT FRONTIER ANALYSIS

Generate 10,000 portfolio configurations to map the feasible set and make the risk-return trade-off explicit for your market.

Each simulated portfolio is evaluated on three dimensions: annualised return (trailing 3 years), annualised volatility, and Sharpe ratio. The resulting scatter plot — the empirical efficient frontier — reveals the risk-return trade-off surface available within this universe. This surface is market-specific: the shape of the SGX frontier differs materially from the KOSPI or ASX frontier due to differences in sector concentration, correlation structure, and individual stock volatility profiles.

PLATFORM INTERFACE



Target Optimization Logic

The tracking optimization algorithm aims to find the optimal subset of stocks that best replicates the performance of our price-weighted index (PWI) with minimal tracking error.

Mathematical Formula

For each simulation iteration, we:

1. **Random Selection**: Randomly select `i` stocks from the universe of `n` dividend-paying stocks
2. **Weight Calculation**: Calculate price-weighted portfolio weights at time `t`:

$$w_{i,t} = \frac{P_{i,t}}{\sum_{j=1}^{i} P_{j,t}}$$

3. **Portfolio Returns**: Calculate tracking portfolio returns:

$$R_{portfolio,t} = \sum_{i=1}^{n} w_{i,t-1} \times R_{i,t}$$

4. **Active Returns**: Compute active returns against the benchmark:

$$R_{active,t} = R_{portfolio,t} - R_{index,t}$$

5. **Tracking Error**: Calculate annualized tracking error:

$$TE = \sigma(R_{active}) \times \sqrt{252}$$

6. **Information Ratio** (optional): Measure risk-adjusted active return:

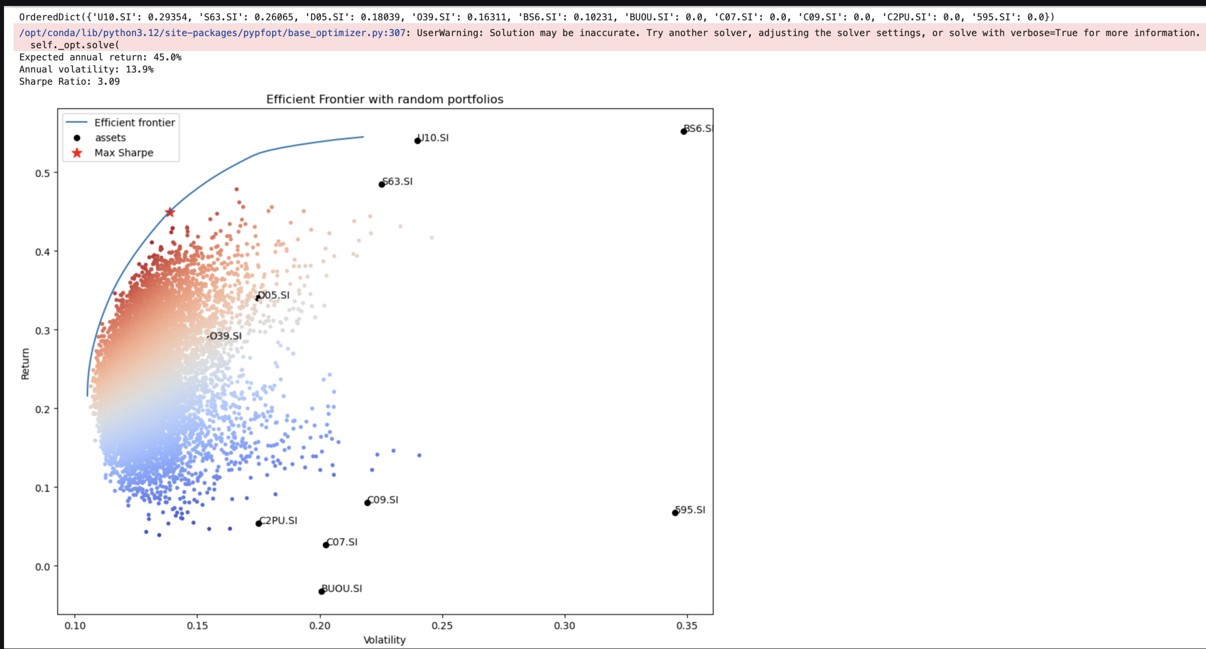
$$IR = \frac{\bar{R}_{active}}{TE}$$

The algorithm runs **10,000 simulations** to find the combination with the lowest tracking error, ensuring optimal index replication with the selected number of stocks. This approach follows industry standards used by major index fund providers like Vanguard and BlackRock.

```
[*]:
# Tracking optimization
const = div_sym.to_list()
n = len(const)
i = IN_STOCK_CNT
sims = 10000
```

Target optimisation logic — the algorithm runs 10,000 simulations to find the optimal portfolio subset. Mathematical formula for weight calculation, portfolio returns, active returns, tracking error, and information ratio are fully documented and reproducible.

PLATFORM INTERFACE



Efficient frontier — SGX equity mandate. Each point is one simulated portfolio. Red = high Sharpe, blue = low Sharpe. Star marks the max-Sharpe selection.

With sufficient sampling the empirical frontier stabilises. 10,000 constrained portfolio configurations are typically adequate to characterise the feasible risk-return region. Modern Portfolio Theory then identifies the maximum Sharpe portfolio from within the mapped space. The frontier plot, with the selected portfolio highlighted, serves as a reproducible artefact for model governance and review.

5. EFFICIENT FRONTIER OPTIMISATION AND STOCK SELECTION

Select on risk-adjusted merit, then validate against the benchmark: combine frontier selection with sector coverage and explicit tracking-error limits.

The final stock selection is validated against the benchmark on two criteria: sector coverage (at least one stock from each of the three primary sectors) and tracking error (annualised tracking error against the STI below 8%). Two diagnostics assess genuine diversification: an allocation-weight view makes concentration transparent, and a correlation matrix confirms that selected holdings are not near-substitutes co-moving through market cycles.

6. GENERALISABILITY AND IMPLEMENTATION

Port the framework by tuning the inputs: set market-appropriate liquidity bars, choose the correct national benchmark, and add diversification constraints when indices are highly concentrated.

Data infrastructure requirements are modest: three years of daily OHLCV price data, dividend history, and daily trading volume — all available from standard financial data providers for most listed markets globally.

6.1 Operationalisation — Platform Workflow

In a production setting, the five-stage method is organised into two phases: (a) strategy construction and validation and (b) client onboarding and allocation. A manager review gate prior to client assignment establishes an approval record for strategy metadata, holdings, weights, and benchmark-validation results.

> Operational completeness requires (i) a funding step with an auditable record and (ii) a defined market-data refresh policy comprising daily OHLCV history with routine updates, supplemented by exchange-rate series for multi-currency mandates.

7. CONCLUSION

In domestic markets, robustness comes from explicit constraints, transparent optimisation, and benchmark-aware validation.

Systematic portfolio construction in domestic equity markets is not merely a scaled-down version of global quant strategies. The constraints of a bounded universe, concentrated sectors, and liquidity limitations demand purpose-built frameworks. The five-stage process described here provides a rigorous, replicable foundation for any single-country equity mandate. The SGX implementation demonstrates that meaningful diversification and strong risk-adjusted returns are achievable even within a relatively small domestic universe.

This paper describes the methodology underpinning the Vektor platform. To see the framework applied live to your mandate — any listed equity market, any currency, any benchmark — enquire about the Founding Mandate Programme at investpuppy.com. For the commercial case, see the companion document: Why Vektor.